



Regular Paper

Chain Reaction: A Triple-Chain Architecture for Sensitive Thumbnail-Preserving Image Encryption

Submission ID 264ad3ce-e07b-4ab1-8005-1dd6f16f6995

Submission Version Initial Submission

PDF Generation 28 Jul 2025 08:16:48 EST by Atypon ReX

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Additional Information

EDICS

IF-CRY Applied cryptography / IF-CRY-MM Multimedia encryption

IF-CRY-MM Multimedia encryption

IF-PRV Anonymization and data privacy / IF-PRV-PROT Privacy protection

IF-PRV-PROT Privacy protection

Subject Category

MULTIMEDIA FORENSICS

WATERMARKING AND STEGANOGRAPHY

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No

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Explain in detail why the contribution of this manuscript is within the scope of the IEEE Transactions on Information Forensics & Security?

This manuscript on the Triple-Chain TPE. The work directly contributes to TIFS's core areas of multimedia security and privacy-preserving cryptography, offering both theoretical advancement (novel chained encryption architecture) and practical protection. Experimental validation confirms our architecture achieves the first sensitive TPE solution, aligning with TIFS's emphasis on empirically grounded security innovations, significantly improving security while maintaining visual usability..

Why is the contribution significant (What impact will it have)?

This work is significant because it fundamentally improves TPE by introducing cryptographic sensitivity. Unlike previous TPE schemes that remain robust against modifications (creating security risks), the triple-chain architecture makes encrypted content noise-sensitive while preserving visual usability. It enables TPE to achieve the cryptographic avalanche effect for the first time, significantly enhancing its applicability in security-sensitive scenarios.

What are the three papers in the published literature most closely related to this paper?

1. Y. Zhang, et al, "A Reversible Framework for Efficient and Secure Visual Privacy Protection," *IEEE TIFS*, 18: 3334-3349, 2023, doi: 10.1109/TIFS.2023.3280341.
2. P. Puteaux et al, "An Efficient MSB Prediction-Based Method for High-Capacity Reversible Data Hiding in Encrypted Images," *IEEE TIFS*, 13(7): 1670-1681, 2018, doi: 10.1109/TIFS.2018.2799381.
3. X. Bi, et al, "Privacy-Preserving Color Image Feature Extraction by Quaternion Discrete Orthogonal Moments," *IEEE TIFS*, 17: 1655-1668, 2022.

What is distinctive/new about the current paper relative to these previously published works?

Regarding the first scheme, it focuses on improving TPE's usability but lacks sensitivity, while our work achieves both. The second scheme employs data hiding for usability but lacks visual perceptibility, which our work preserves. The third scheme prioritizes image security without considering usability, whereas our encrypted images preserve usability.

Files for peer review

All files submitted by the author for peer review are listed below. Files that could not be converted to PDF are indicated; reviewers are able to access them online.

Name	Type of File	Size	Page
bare_jrnl.pdf	Main Document - PDF	2.9 MB	Page 5

Chain Reaction: A Triple-Chain Architecture for Sensitive Thumbnail-Preserving Image Encryption

Ruoyu Zhao, Wenying Wen, William Puech, Rushi Lan, and Yushu Zhang

Abstract—Recently, thumbnail-preserving encryption (TPE) has attracted significant attention, as it maintains certain visual information in encrypted images, thereby achieving a trade-off between visual usability and privacy protection. However, existing TPE schemes operate with independent encryption between pixels, blocks, or channels, which results in excessive robustness and consequently introduces security vulnerabilities. To address this, we propose a Triple-Chain Architecture to achieve sensitive TPE. This architecture integrates three interdependent encryption chains at the pixel, block, and channel levels, establishing cryptographic dependencies. Any minor modification to either the original or encrypted image results completely divergent encryption/decryption outcomes. Specifically, the architecture introduces three chained encryption mechanisms: (1) Pixel-level chained encryption propagates encrypted pixels outputs as the part of subsequent inputs; (2) Block-level hashing derives new keys from plaintext-key concatenations; (3) Channel-level dependencies are established via iterative key derivation. This cascading structure ensures cryptographic interdependency across all levels. The experimental results verify that the proposed architecture achieves sensitive TPE for the first time while preserving the TPE characteristics of visual usability and privacy protection.

Index Terms—Image encryption, privacy, visual usability, multimedia security.

I. INTRODUCTION

IN the information age, images have become a crucial medium of information due to capacity to convey rich and authentic content, leading to their widespread use and dissemination. Proverbs across cultures, such as “a picture is worth a thousand words” [1], powerfully demonstrate the immense informational capacity, perceived reliability, and cognitive simplicity of images in human perception. Therefore, images are extensively utilized across diverse domains, including social media [2], life documentation [3], and scientific research [4]. However, while images are widely adopted due to their advantages, they also pose significant privacy security risks [5].

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Compared to textual data, images allow viewers to readily perceive (or believe they perceive) authentic privacy information, which can then propagate rapidly through the digital networks. The 2014 iCloud celebrity photo leaks [6] stand as one of the most emblematic digital privacy breaches of the information age. A vast quantity of privacy images rapidly disseminated across the global internet within hours. This incident demonstrated with unprecedented clarity the catastrophic consequences when privacy images intersects with digital connectivity. Meanwhile, in 2024, Alibaba Cloud Drive experienced a photo exposure incident [7] where users’ cloud-stored images became accessible in other users’ accounts, leading to large-scale privacy breaches.

Traditional image security methods include watermarking, filtering, and encryption. Watermarking embeds identifiers into images to indicate source, copyright, and prevent unauthorized copying [8], but the original visual content remains exposed. Filtering, e.g., mosaic or blur, removes details irreversibly, making it unsuitable for scenarios requiring image recovery. Encryption is a reversible countermeasure against image breaches [9], [10]. However, these schemes typically transform visually meaningful images into encrypted images that resemble salt-and-pepper noise, thereby completely eliminating their visual usability [11]–[13]. Although the encryption method appears intuitive, it introduces significant practical limitation. For instance, encrypted images become visually indistinguishable from others, and without decryption, human cannot discern any meaningful content within them. Furthermore, if users need to locate a specific image, they should blindly decrypt all encrypted images one by one until finding the desired one, which is an inefficient process.

Recently, researchers have proposed the concept of thumbnail-preserving encryption (TPE) [14], which effectively balances image privacy protection with visual usability. Specifically, the image is divided into multiple square blocks to be encrypted by sum-preserving encryption, resulting in encrypted images that preserve visual consistency with the thumbnails. In TPE, intra-block visual information, treated as sensitive, is obscured, while inter-block information, considered privacy-irrelevant, is preserved.

Upon its introduction, TPE immediately garnered widespread attention and has since undergone extensive improvements from various perspectives, resulting in numerous proposed schemes [15]. Currently, the majority of existing TPE variants are derived from the foundational scheme [16]. This scheme demonstrates that when employing a substitution-permutation encryption architecture, pixels within each block should be grouped in pairs, with substitution

encryption applied to each pixel pair. It effectively resolves the implementation challenges of achieving sum-preserving substitution encryption at the whole-block level.

Motivated by this, in this paper, we propose a sensitive TPE architecture where the encryption of pixel groups, blocks, and channels is interlinked, in which each step's outcome directly and significantly influences the next. While pixel groups remain the basic encryption unit, the process features triple-chain dependencies. First, pixel-level chaining, the last encrypted pixel and one unencrypted pixel feed into the next substitution encryption; Second, block-level chaining, each block's key is dynamically derived from the prior block's data and key; Third, channel-level chaining, cross-channel encryption initializes each subsequent channel's key from the preceding channel's final block and derived key. This design creates a cryptographic chain reaction, akin to nuclear physics processes, where minor input perturbations result exponentially amplified effects. In all, the main contributions of this work are:

- We propose a novel Triple-Chain TPE architecture, which realizes the first sensitivity-aware TPE scheme where any minute alteration in the image propagates globally during both encryption and decryption processes;
- We propose a three-level chained encryption structure (pixel-level, block-level, and channel-level), where the interdependencies among pixels, blocks, and channels during encryption create an avalanche effect. Any modification at any level significantly impacts subsequent encryption results;
- Experimental results show that the proposed architecture achieves strong sensitivity while maintaining TPE properties: altering just one element in the encrypted image ensures that no additional information is leaked beyond the encrypted one after decryption, while single original element changes produce completely different encryption results.

The remainder of this paper is organized as follows. Section II describes the previous image encryption schemes about traditional encryption and TPE. Section III presents the construction of the proposed architecture. Experiments are listed in Section IV, and this paper is concluded in Section V.

II. RELATED WORK

In this paper, some previous methods on traditional image encryption and TPE are introduced.

A. Traditional Image Encryption

Traditional image encryption constitutes a well-established research domain that holds critical significance in multimedia security. Fundamentally, it achieves security through two primary mechanisms: statistical feature destruction and visual perception obfuscation, i.e., indistinguishability. From a technical implementation perspective, existing schemes seldom employ bitstream encryption algorithms (e.g., AES and elliptic curve cryptography) due to their inherent limitations in format compatibility and computational efficiency [17]. Currently, many image encryption schemes employ technologies such as

chaotic systems [18], optical encryption [19], DNA encoding [20], and compresses sensing [21] in their architectural designs. However, they also render the image unusable prior to decryption.

Several research initiatives have proposed innovative workarounds to maintain partial image usability in encrypted states. For example, encrypted image data hiding [22], [23], it enables textual read/write operations directly on ciphertext; Encrypted image retrieval [24], it preserves controllable feature descriptors for content-based search. However, the usability of these schemes fundamentally stems from non-visual information components, which inherently contradicts the intuitive way by obtaining information through vision [25].

A subset of researchers has attempted to mitigate this challenge through selective encryption or region-of-interest (RoI) encryption strategies [26], [27]. Specifically, these schemes divide images into privacy-sensitive and non-sensitive areas; the sensitive ones are encrypted, which non-sensitive ones remain unprocessed. However, these schemes rely on overly idealized assumptions, in which sensitive regions can be clearly distinguished. In reality, sensitive information can often be inferred from seemingly non-sensitive parts. For example, while a person's face in an image is directly linked to their identity, their clothing may also indirectly reveal identifiable information [28].

Several other block-based image encryption schemes [29], [30] also exhibit distinct block-like visual effects in the encrypted images. However, these visual characteristics are merely incidental by-products of their efficiency-focused design philosophy. These schemes are not originally intended to preserve visual quality, focusing solely on security considerations, which consequently results in suboptimal visual performance.

B. Thumbnail-Preserving Encryption

TPE, initially proposed in 2015 [14], employs undifferentiated encryption across all image regions. While this process eliminates detailed visual information, it deliberately preserves coarse-grained visual data. Crucially, the block size parameter can be readily adjusted to modulate the amount of residual visual information in encrypted images, thereby balancing between visual usability and privacy protection. However, this scheme is based on permutation-only encryption, which has been demonstrated by numerous cryptanalytic studies to contain inherent security vulnerabilities [31].

In 2019, Tajik *et al.* [16] innovatively proposed a novel architecture that achieved TPE under a permutation-substitution framework by further dividing block elements into pixel groups and performing sum-preserving substitution encryption on these groups. Subsequently, several schemes [32] attempted to improve this scheme by increasing the number of pixels in groups. Some researchers have explored combining this architecture with data hiding techniques to enhance usability. For example, Zhao *et al.* [15] proposed compressing the most significant bits to create reversible embedding space for additional data, thereby providing non-visual usability. In all, these schemes are based on the architecture [16],

either designing new encryption methods for pixel groups or transplanting other techniques, e.g., data hiding.

Currently, TPE schemes exhibit robustness. First, schemes decompose color images into separate channels, processing each channel completely independently. Second, each channel is further divided into multiple blocks, where the encryption of each block remains independent from others. Third, whether employing permutation-only encryption or group substitution encryption within blocks, current implementations demonstrate robustness.

The left portion of Fig. 1 illustrates the schematic architecture of previous schemes. It ensures that pixel corruption only affects elements within the group without outward propagation, thereby granting the encrypted images strong robustness. However, such design proves unsuitable for security-sensitive scenarios [33]. On the other hand, the proposed triple-chain architecture, as shown in the right portion of Fig. 1, interconnects pixels, blocks, and channels, enabling pixel-level errors to propagate throughout the entire image.

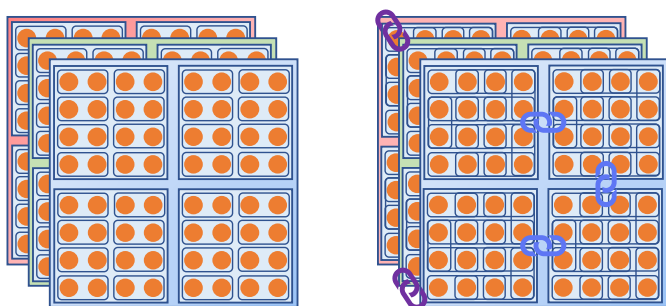


Fig. 1: The basic architecture of TPE. Left: the previous ones; Right: the proposed one.

III. THE PROPOSED ARCHITECTURE

The above analysis reveals that existing TPE schemes lack sensitivity considerations, which motivates our work. We propose a triple-chain encryption architecture, where it enables TPE to achieve the *avalanche effect* for the first time. For encryption, even a minor difference in the original image produces entirely distinct encrypted images; For decryption, a slight modification to the encrypted image makes the decrypted result unrecognizable, effectively preventing successful recovery.

The overall encryption process is shown in Fig. 2. A color image consists of three independent channels, each being a 2D integer matrix. First, *dividing blocks*: each channel is divided into blocks with size $b \times b$ as needed, consisting with previous TPE schemes. Second, *pixel-level chaining encryption*: one element from the previously encrypted pixel group combines with an unencrypted pixel to form a new pixel group for encryption. The initialization key $K_{1,1,1}$ held by the user is applied to the pixels in the first block. Third, *block-level chaining encryption*: after encrypting a block, the key and the block's original pixel values are combined to derive a new key for encrypting the next block. Fourth, *channel-level chaining encryption*: after encrypting a channel, the original elements in the last block are combined with the derived key

to drive a new key for encrypting the next channel. Finally, the encrypted image is obtained once all elements are encrypted. The following is a detailed introduction to the process of this architecture.

A. Dividing Blocks

The image I can be represented:

$$I \in \{0, 1, \dots, 255\}^{w \times h \times c}, \quad (1)$$

where w and h mean the width and height of the image, respectively, and $c = 3$, which presents the channel numbers (R, G, and B).

The block size parameter b is required when dividing the image into blocks, meaning the image is partitioned into multiple blocks with the size $b \times b$. I can be represented:

$$I = \{B_{i,j,k} | 1 \leq i \leq \frac{w}{b}, 1 \leq j \leq \frac{h}{b}, 1 \leq k \leq 3\}, \quad (2)$$

where $B_{i,j,k}$ denotes a block, with i , j , and k representing the block indices. In this work, it is assumed that b can exactly divide both w and h .

B. Pixel-Level Chained Encryption

In this part, sum-preserving encryption of elements in the block is implemented through two major steps: substitution and permutation. This work requires a 256-bit secret key K and a pseudorandom function generator (PRNG) $G_K(\eta)$:

$$\begin{aligned} K_{1,1,1} &\in \{0, 1\}^{256} \\ \vec{P} &\leftarrow G_K(\eta) \end{aligned}, \quad (3)$$

where $K_{1,1,1}$ is completely uniformly chosen bitstream, $G_K(\eta)$ means the PRNG driven by the secret key K and outputs η pseudorandom numbers, and $\vec{P}$ denotes the vector composed of the η elements output by $G_K(\eta)$. In this work, the PRNG designed by Hua *et al.* [34] is employed as $G_K(\eta)$, which has been extensively experimentally verified to produce output distributions computationally indistinguishable from true random numbers. It should be noted that users can replace it with their PRNG as needed, which does not affect this architecture.

1) Substitution

In previous TPE schemes, substitution encryption independently processed pixel groups within blocks, resulting in intra-block robustness. To address this limitation, we propose pixel-level chained substitution encryption in this work. While still operating on two elements per encryption, the groups are no longer independent. Instead, the second element from the previously encrypted group is paired with a new plaintext element to form the next encryption group. The pixel-level substitution encryption logic of the proposed architecture and a concrete instance are illustrated in Fig. 3.

In this part, this substitution encryption of pixel groups is based on sum-preserving encryption of the element values within each group. Given a pixel group $\vec{g} = (p_1, p_2)$, the encryption process is:

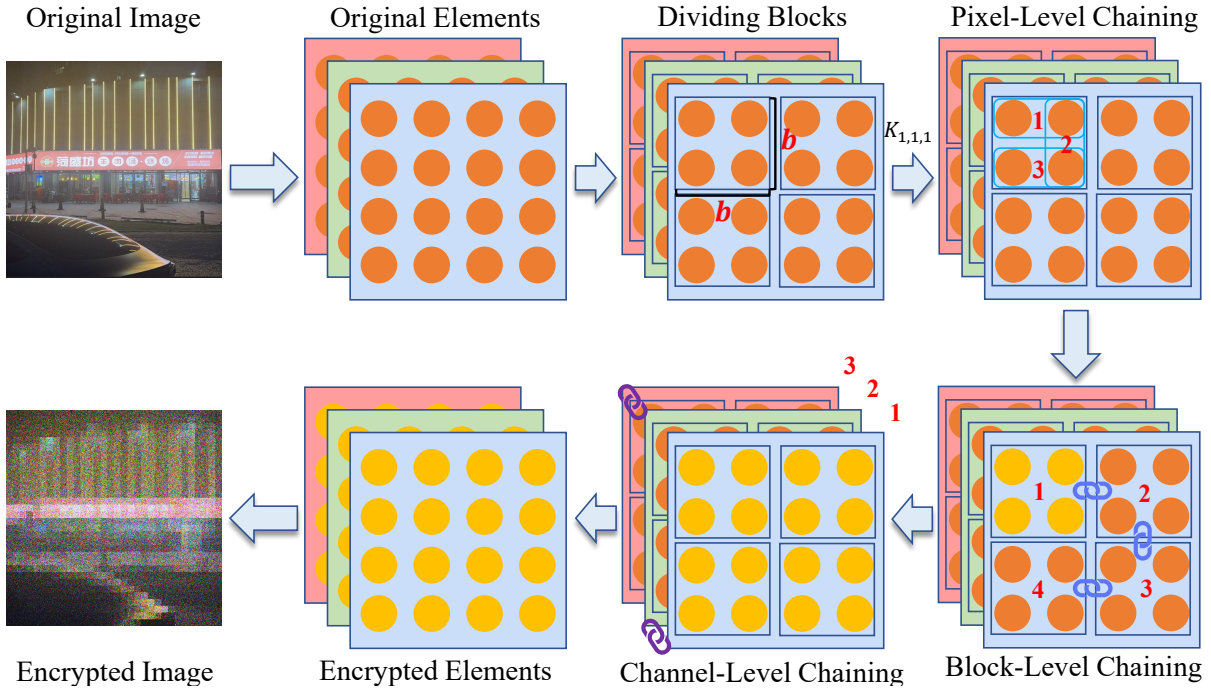


Fig. 2: Overall encryption process in the architecture, where the numbers represent the order of encryption at each level, orange represents the original element, and yellow means the encrypted element.

- 1) The sum of the pixel group $\vec{g}$ is:

$$s = p_1 + p_2; \quad (4)$$

- 2) A set $\Phi(s)$ is constructed by enumerating all admissible pixel groups that share the same sum s of element values:

$$\Phi(s) = \{(x, y) \in \{0, 1, \dots, 255\}^2 | x + y = s\}; \quad (5)$$

- 3) Each vector (i.e., pixel group) is assigned sequential indices starting from 0 in $\Phi(s)$:

$$\Phi(s) = \{\vec{g}_0, \vec{g}_1, \dots, \vec{g}_{|\Phi(s)|-1}\}, \quad (6)$$

where $g_i \neq g_j$ for all $i \neq j$, $|\cdot|$ means the number of elements in the set, and $|\Phi(s)|$ can be computed:

$$|\Phi(s)| = \begin{cases} s + 1, & s \leq d \\ 2 \times d - s + 1, & \text{otherwise} \end{cases}, \quad (7)$$

where $d = 255$;

- 4) The index of a pixel group in the set $\Phi(s)$ can be directly computed:

$$\text{rank}(\vec{g}) = \begin{cases} p_1, & s \leq d \\ p_1 - s + d, & \text{otherwise} \end{cases}; \quad (8)$$

- 5) For a pixel group with index r , its encryption process proceeds:

$$r_e = r + \xi \bmod |\Phi(s)|, \quad (9)$$

where ξ represents a random number, mod means the remainder operation;

- 6) Based on Eq. (8), the pixel group can be inversely derived from its index:

$$\text{rank}_s^{-1}(r) = \begin{cases} (r, s - r), & s \leq d \\ (s - d + r, d - r), & \text{otherwise} \end{cases}. \quad (10)$$

Therefore, r_e can be input into Eq. (10) to obtain an encrypted pixel group $\vec{c}$ with the identical element value sum s .

The aforementioned encryption process is summarized as group substitution encryption, denoted by the function $E_{se}(\vec{g}, \xi)$. The following provides a more detailed process of pixel group encryption. Formally, each block B is represented:

$$B = \begin{bmatrix} p_{1,1} & p_{1,2} & \cdots & p_{1,b} \\ p_{2,1} & p_{2,2} & \cdots & p_{2,b} \\ \vdots & \vdots & \ddots & \vdots \\ p_{b,1} & p_{b,2} & \cdots & p_{b,b} \end{bmatrix}, \quad (11)$$

The block B contains a total of $b \times b$ elements, which should be divided into $b \times b - 1$, called num_g . The PRNG $G_K(\cdot)$ should iterate num_g times under the key K to output num_g numbers:

$$\vec{P}_{se} = (\delta_1, \delta_2, \dots, \delta_{num_g}), \vec{P}_{se} \in G_K(num_g). \quad (12)$$

It should be noted that the PRNG constructed in the work [34] outputs values that are very small in magnitude and may be positive or negative, thus requiring processing:

$$\xi_i = \lfloor \sqrt{\delta_i^2} \times 10^6 \rfloor, i = \{1, 2, \dots, num_g\}, \quad (13)$$

where $\lfloor \cdot \rfloor$ means the round down, the multiplication by 10^6 serves to amplify the numerical values, and users can adjust this parameter as needed. $\vec{P}_{se}$ can be converted as $\vec{P}_\xi = \{\xi_1, \xi_2, \dots, \xi_{num_g}\}$. Then, the elements in the block B can undergo chained substitution encryption as described in Algorithm 1. The encryption sequence of elements within

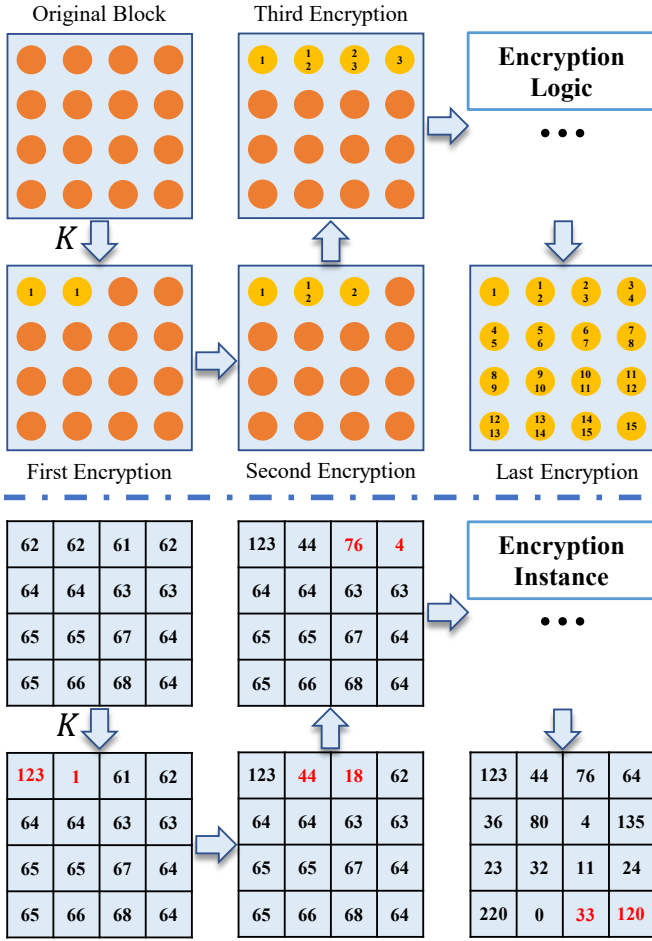


Fig. 3: An example of the proposed pixel-level chained substitution encryption in a $b \times b$ block. Upon: Encryption logic, where numbers indicate in which encryption the element is encrypted; Bottom: Encryption instance, where red represents the elements used in this encryption.

the block is illustrated in the upper of Fig. 3, while a concrete instance is demonstrated in the bottom of Fig. 3.

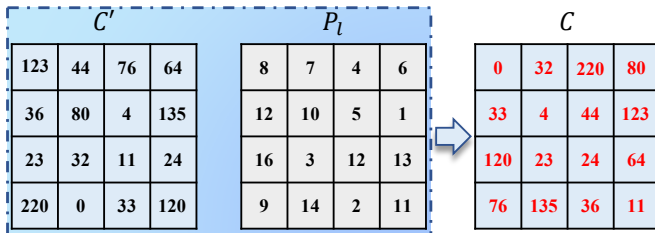


Fig. 4: An example of the proposed pixel-level permutation encryption.

2) Permutation

After performing substitution encryption to obtain C' , a permutation encryption is then applied to its elements, which still requires random numbers generated by the PRNG. Notably, the PRNG employed in [34] outputs two random numbers per iteration: one is used for substitution encryption, while the other is reserved for permutation encryption.

Algorithm 1: Chained Substitution Encryption in B

Input: B , $\vec{P}_\xi$, and num_g
Output: C'
 $\vec{B} \leftarrow Blo2Vec(B)$ // $Blo2Vec(\cdot)$ transforms a two-dimensional block into a one-dimensional vector by concatenating its rows in sequential order
for $i \leftarrow 1 : 1 : num_g$ **do**
 $flag_{begin} \leftarrow i$
 $flag_{end} \leftarrow i + 1$
 $\vec{g} \leftarrow \vec{B}(flag_{begin} : flag_{end})$ // Elements from position $flag_{begin}$ to $flag_{end}$ in $\vec{g}$ are extracted to construct the pixel group
 $\xi_i \leftarrow \vec{P}_\xi(i)$
 $\vec{c} \leftarrow E_{se}(\vec{g}, \xi_i)$
 $\vec{B}(flag_{begin} : flag_{end}) \leftarrow \vec{c}$ // The encrypted group is placed back into the vector to be encrypted
end
 $C' \leftarrow Vec2Blo(\vec{B})$ // $Vec2Blo(\cdot)$ transforms a vector back into a matrix, serving as the inverse of $Blo2Vec(\cdot)$
return C'

Since B contains $b \times b$ elements, the PRNG $G_K(\cdot)$ should iterate $b \times b$ times to generate $b \times b$ random numbers for the permutation operation:

$$\vec{P}_{pe} = (\gamma_1, \gamma_2, \dots, \gamma_{b^2}), \vec{P}_{pe} \in G_K(b^2). \quad (14)$$

Every element in $\vec{P}_{pe}$ has a unique value since all values generated by the PRNG are distinct, i.e.:

$$\gamma_i \neq \gamma_j \text{ for all } i \neq j \in \{1, 2, \dots, b^2\}. \quad (15)$$

The ascending-order indices for elements in $\vec{P}_{pe}$ are computed, and this process is listed in Algorithm 2.

Algorithm 2: Ascending-Order Indices

Input: $\vec{P}_{pe}$
Output: $\vec{P}_l$
 $\vec{P}_l \leftarrow \text{zeros}(b^2)$ // Let $\vec{P}_l$ be initialized as a zero vector of length b^2
 $t \leftarrow 1$
while $\vec{P}_{pe} \neq \emptyset$ **do**
 $\gamma_i \leftarrow \text{IndMin}(\vec{P}_{pe})$ // $\text{IndMin}(\cdot)$ means the index of the minimum element in $\vec{P}_{pe}$
 $\vec{P}_l(i) \leftarrow t$ // The index of the i -th element is t
 $t = t + 1$
 $\vec{P}_{pe} \leftarrow (\vec{P}_{pe}(1), \vec{P}_{pe}(2), \dots, \vec{P}_{pe}(i-1), \vec{P}_{pe}(i+1), \dots, \vec{P}_{pe}(b^2))$ // The indexed element is removed
end
return $\vec{P}_l$

The output of Algorithm 2 is a one-dimensional vector $\vec{P}_l$, which is then reshaped into a two-dimensional matrix of

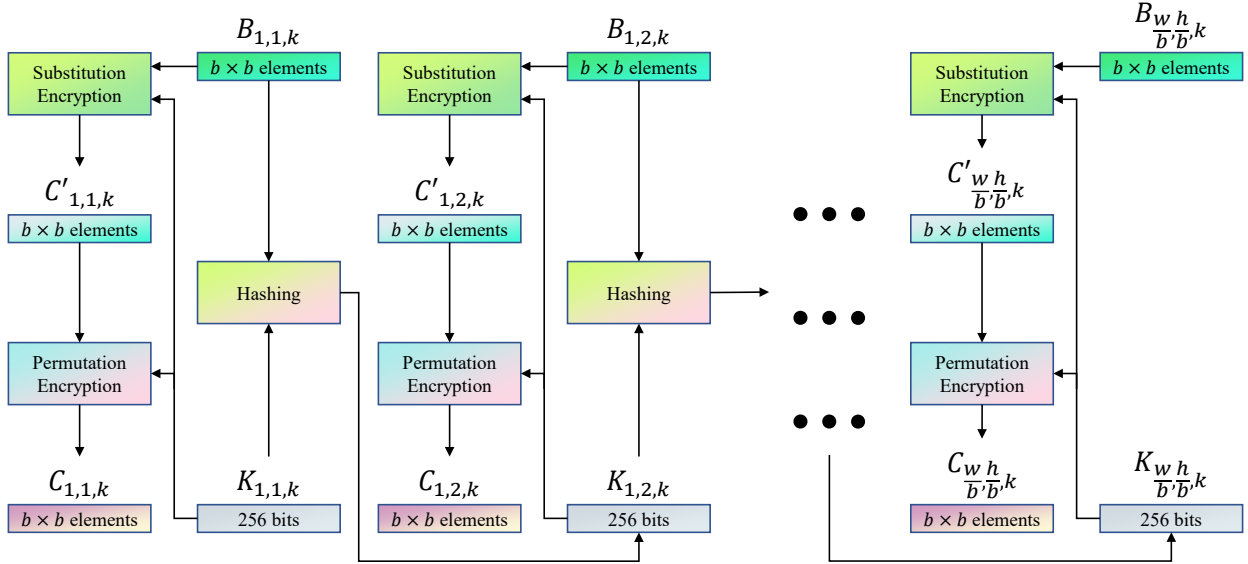
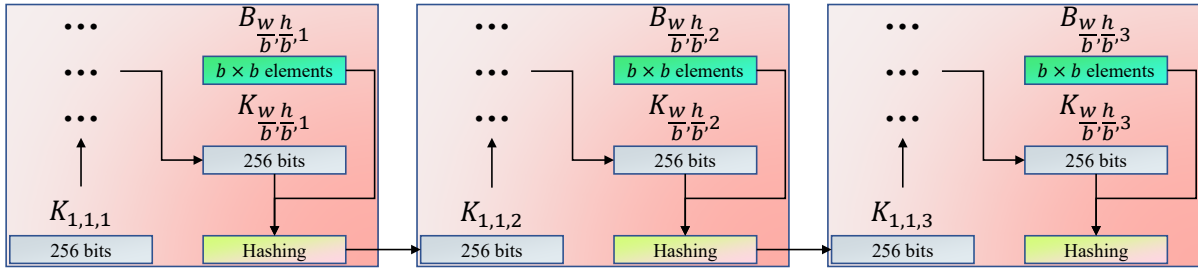
Fig. 5: The overall process of the proposed block-level chained encryption in a channel k .

Fig. 6: The overall process of the proposed channel-level chained encryption.

the same size as the block B , called P_l . Then, the elements in the substituted-encryption block C' undergo permutation encryption based P_l to get the encrypted block C . Specifically, the value $P_l(i, j)$ indicates the ordinal position (in column-major order) of $C(i, j)$ within C' :

$$\begin{aligned} x_{sub} &= \lceil \frac{P_l(i, j)}{b} \rceil \\ y_{sub} &= P_l(i, j) - (\lceil \frac{P_l(i, j)}{b} \rceil - 1) \times b, \\ C(i, j) &= C'(x_{sub}, y_{sub}) \\ P_l(i, j) &\in \{1, 2, \dots, b^2\} \end{aligned} \quad (16)$$

where $\lceil \cdot \rceil$ means round up to an integer. A specific example of permutation encryption is shown in Fig. 4.

C. Block-Level Chained Encryption

The steps constitute the intra-block (pixel-level) encryption process. In previous TPE schemes, all blocks independently encrypt their internal elements without mutual interference, which leads to the following critical limitations in sensitivity. Even if significant modifications occur within a single block, the effects remain confined to that block itself and cannot propagate to other blocks.

In this work, we propose the *key chaining* concept, where the encryption key for each block is derived from both the previous key and the original block's content:

$$\begin{cases} K_{i,j+1,k} = H(K_{i,j,k} || B_{i,j,k}), & \frac{j \times b}{h} \bmod 1 \neq 0 \\ K_{i+1,1,k} = H(K_{i,j,k} || B_{i,j,k}), & \text{otherwise} \end{cases}, \quad (17)$$

where $H(\cdot)$ means the a cryptographic hash function that accepts inputs of arbitrary length and outputs a fixed 256-bit digest, and $||$ represents the concatenation operator. In the key derivation process, the deliberate choice to use the original data of the previous block (rather than its ciphertext) ensures that keys cannot be derived directly from ciphertext during decryption.

The security can be improved from two aspects using the key chaining:

First, cross-block dependency enhances security:

- Key chaining mechanism encrypts all blocks into inter-dependent ciphertext units;
- Modification of any block triggers a cascading effect;
- Attackers cannot perform localized tampering, and they need compromise the entire chained structure.

Second, each blocks' encryption key is essentially a dynamically generated 256-bit pseudorandom key, exhibiting a one-time-pad-like property [35]:

- Identical original blocks yield completely different ciphertexts at different positions;

- A single-pixel difference causes downstream keys to change entirely;
- Even with the same initial key, encrypting different images produces a one-time-pad-like effect.

Fig. 5 illustrates the block-level chained encryption process within the k -th channel. As demonstrated, the encryption key for each block is derived by combining the previous block's key and the original block.

D. Channel-Level Chained Encryption

In previous TPE schemes, encryption/decryption is channel-independent, meaning operations in one channel affect nor affected by other channels. In this work, the aforementioned two-level chained encryption (pixel-level and block-level) still fails to address the channel-independent problem. In this part, we introduce channel-level key chaining to deeply couple the encryption processes across different channels.

Fig. 6 shows the channel-level chained encryption process proposed in this work. A detailed description of this procedure is then provided.:

- 1) The first block of the first channel $B_{1,1,1}$ is encrypted using the user-provided initial key $K_{1,1,1}$;
- 2) All subsequent blocks within this channel are encrypted using derived keys (as shown in Eq. (17)) until the final block $B_{\frac{w}{b}, \frac{h}{b}, 1}$ is processed;
- 3) The final original block $B_{\frac{w}{b}, \frac{h}{b}, 1}$ of the channel and its corresponding key $K_{\frac{w}{b}, \frac{h}{b}, 1}$ are combined to derive the encryption key $K_{1,1,2}$ for the first block $B_{1,1,2}$ of the next channel:

$$K_{1,1,2} = H(K_{\frac{w}{b}, \frac{h}{b}, 1} || B_{\frac{w}{b}, \frac{h}{b}, 1}); \quad (18)$$

- 4) Blocks in the second channel are encrypted following the same procedure as Step 2);
- 5) The key required for the first block $K_{1,1,3}$ of the third channel is derived in the same manner as Step 3), and its internal blocks are encrypted identically to Step 2).

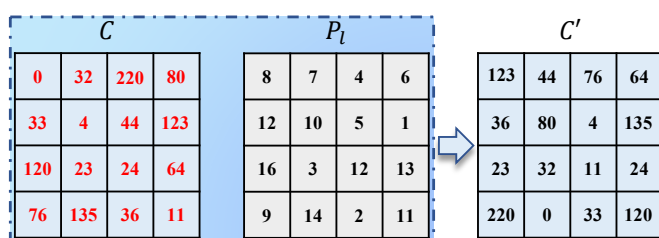


Fig. 7: An example of the proposed pixel-level permutation decryption.

E. Encryption Round

In encryption schemes, iterative encryption is frequently employed to enhance security by applying encryption algorithms to data multiple times. Each complete execution of the encryption algorithm is referred to as an encryption round.

In this work, multi-round encryption can similarly be employed to enhance the encrypted image security. However,

Algorithm 3: Chained Substitution Decryption in C'

Input: C' , $\vec{P}_\xi$, and num_g
Output: B
 $\vec{C}' \leftarrow Blo2Vec(C')$
for $i \leftarrow 1 : 1 : num_g$ **do**
 $flag_{begin} \leftarrow i$
 $flag_{end} \leftarrow i + 1$
 $\vec{c} \leftarrow \vec{C}'(flag_{begin} : flag_{end})$
 $\xi_i \leftarrow \vec{P}_\xi(i)$
 $\vec{g} \leftarrow D_{se}(\vec{c}, \xi_i)$
 $\vec{C}'(flag_{begin} : flag_{end}) \leftarrow \vec{g}$
end
 $B \leftarrow Vec2Blo(\vec{C}')$
return B

to further strengthen sensitivity protection, the alternating encryption sequence is implemented:

- 1) Forward Phase. In the odd-numbered round, the encryption process executes sequentially from Channel 1 to Channel 3;
- 2) Backward Phase. In the even-numbered round, it is executed in reverse channel order, i.e., Channel 3 to Channel 1.

This alternating forward-backward encryption sequence ensures that any modifications in the image can propagate throughout the entire image during the encryption/decryption process.

F. Decryption

During decryption, the process should follow the reverse sequence of forward and backward rounds as encryption. While the channel processing order differs between directions, the decryption procedure itself remains identical. Overall, the decryption process in this work is the inverse of the encryption process described. We detail only the decryption of the first block $C_{1,1,1}$ in the first channel, as the remaining encrypted blocks and channels can be decrypted sequentially following the same logic.

The encrypted image E can be represented:

$$E = \{C_{i,j,k} | 1 \leq i \leq \frac{w}{b}, 1 \leq j \leq \frac{h}{b}, 1 \leq k \leq 3\}. \quad (19)$$

For $C_{1,1,1}$, it first undergoes permutation decryption, followed by pixel-level chained decryption.

An example of permutation decryption is shown in Fig. 7, in which the generation manner of P_l is listed in Algorithm 2. The value $P(i, j)$ means the target position (in column-major order) to which element $C(i, j)$ should be decrypted:

$$\begin{aligned} x_{sub} &= \lceil \frac{P_l(i, j)}{b} \rceil \\ y_{sub} &= P_l(i, j) - (\lceil \frac{P_l(i, j)}{b} \rceil - 1) \times b. \\ C'(x_{sub}, y_{sub}) &= C'(i, j) \\ P_l(i, j) &\in \{1, 2 \dots, b^2\} \end{aligned} \quad (20)$$

Then, the process of pixel-level chained substitution decryption is:

- 1) $\vec{P}_\xi = \{\xi_1, \xi_2, \dots, \xi_{num_g}\}$ is generated as Eq. (13);
- 2) For the decryption process $D_{se}(\vec{g}, \xi_i)$ of the encrypted pixel group $\vec{c}$, it is generally the same as $E_{se}(\vec{g}, \xi_i)$, but Eq. (9) becomes:

$$r = r_e - \xi \bmod |\Phi(s)|, \quad (21)$$

- 3) $C'_{1,1,1}$ is input into Algorithm 3 for substituting encryption to get the original block $B_{1,1,1}$.

Once an encrypted block is decrypted to its original data, the original block and its key are combined to derive a new key for decrypting the next block. This process repeats sequentially until all block in $E = \{C_{i,j,k} | 1 \leq i \leq \frac{w}{b}, 1 \leq j \leq \frac{h}{b}, 1 \leq k \leq 3\}$, ultimately recovering the original image I .

IV. EXPERIMENTAL RESULTS

In this section, the presented experimental results are divided into three parts: qualitative results, key security analysis, and comparisons with previous work.

Unless otherwise specified, all experiments use images of size 512×512 , with 256-bit secret key $K_{1,1,1}$:

$$K_{1,1,1} = 357538782F413F4428472B4B6250655368566D59703373367639792442264529. \quad (22)$$



Fig. 8: Examples of five images encrypted by the proposed method under the different block sizes.

A. Qualitative Results

In this part, 5 daily-life images captured by the authors, $I_i = \{1, 2, \dots, 5\}$, are encrypted using the proposed work, with block sizes $2^i, i = \{3, 4, 5, 6\}$. In subsequent experiments, these five images are also selected for testing.

The obtained results of the encrypted images are presented in Fig. 8. As observed, the encrypted images preserve rough visual information, with smaller blocks preserving more information and larger blocks preserving less. The detailed information within blocks is hidden and protected, while the visual information between blocks is preserved. Consequently, this work achieves a tunable trade-off between usability and privacy through straightforward adjustment of the block size parameter. Moreover, the overall macroscopic color perception preserves largely consistent with the original image. The reason lies in the sum-preserving encryption applied to each block's internal information, where the sum of element values preserves unchanged after encryption. This aligns with thumbnail generation methods, thereby preserving visual information.

	I_1	I_2
$K_{1,1,1}$	357538782F413F4428472B4B6250655368566D59703373367639792442264529	
$K_{1,2,1}$	75D689451BC7A26645A43BEE78A0D63F1791E60A89D0E601AA5DDA95DBBB652F	ADA8152848D0E0C309CB0D4392A3C5734516C29A0B1F19734297311975059B5B
$K_{1,1,2}$	C48E26C1A10B59FF189EDB5A327D569115A0B688ED072CE73BF15752DD8B5D05	E4616BFDD51872CC2B96E274BD4650AB7C4C27523A80406EAAFA48E0927BE4847
$K_{1,1,3}$	09D089E66206C5F07D308612C381827E8A32B8D8680001DAA560215DCFE4AB8D	C25F9A13BFB1891134CE70EE5359498EACC361E94646CAB98766B6D0415334ED

Fig. 9: An example of the derived keys obtained by encrypting two images I_1 and I_2 with the same initial key $K_{1,1,1}$, $b = 32$.

	$b = 32$	$b = 64$
$K_{1,1,1}$	357538782F413F4428472B4B6250655368566D59703373367639792442264529	
$K_{1,2,1}$	75D689451BC7A26645A43BEE78A0D63F1791E60A89D0E601AA5DDA95DBBB652F	E4841B626173CA15F430CA9BF013AFDC674458612C992AB183690A8A7D0CADE7
$K_{1,1,2}$	C48E26C1A10B59FF189EDB5A327D569115A0B688ED072CE73BF15752DD8B5D05	DBB55AF9E60D4D3A008A876543FD4273611E4EE6AA2B9FBEF92D4D5B1BB9BC63
$K_{1,1,3}$	09D089E66206C5F07D308612C381827E8A32B8D8680001DAA560215DCFE4AB8D	D754AEDDFFDEB82D2F265D9FFD90C850EA3A3E2C23F381347826295C3931AA43

Fig. 10: An example of the derived keys obtained by using different block sizes, $b = 32$ and $b = 64$, with the same initial key $K_{1,1,1}$.

	I_1	I'_1
$K_{1,1,1}$	357538782F413F4428472B4B6250655368566D59703373367639792442264529	
$K_{1,2,1}$	75D689451BC7A26645A43BEE78A0D63F1791E60A89D0E601AA5DDA95DBBB652F	122D2FEA30DD54F49C6CE4B6CE6705BB470012A6BF458D1BFB86FF1FF3E40BB8
$K_{1,1,2}$	C48E26C1A10B59FF189EDB5A327D569115A0B688ED072CE73BF15752DD8B5D05	1AFF1426D8FEBBAA513136D4C8A4829A212E42D10B4BCE45ACDB0B4A0D568C7B
$K_{1,1,3}$	09D089E66206C5F07D308612C381827E8A32B8D8680001DAA560215DCFE4AB8D	B9F16213B268729A9A237E49CCB70EBC65A30390D51C1A1DED3A448A8A028BE7

Fig. 11: An example of the derived keys obtained by encrypting two images I_1 and I'_1 with the same initial key $K_{1,1,1}$, $b = 32$.

B. Key Security Analysis

In this part, the security of the proposed architecture is evaluated from three aspects, key space, derivation security, and key sensitivity.

	I_1	
$K_{1,1,1}$	357538782F413F4428472B4B62506553 68566D59703373367639792442264529	357538782F413F4428472B4B62506553 68566D59703373367639792442264528
$K_{1,2,1}$	75D689451BC7A26645A43BEE78A0D63F 1791E60A89D0E601AA5DDA95DDBB652F	66A0B36AE432AFD9E09D9141E5BEAD4B D988537BAC8B403C6648E5FABEF2FCBA
$K_{1,1,2}$	C48E26C1A10B59FF189EDB5A327D5691 15A0B688ED072CE73BF15752DD85D05	AD814438C52D118DC2575C7CADEC89F8 23914BEE4846E3AFF157913BC884EFF5B
$K_{1,1,3}$	09D089E66206C5F07D308612C381827E 8A32B8D8680001DAA560215DCFE4AB8D	99D7092A6BDF7584B8ABC5F2AC16C4C 91DCB3CCA4BF191ECFF5C8260B96AE77

Fig. 12: An example of the derived keys obtained by two initial keys that differ by 1 bit, $b = 32$.

1) Key Space Analysis

The key space represents the resistance against brute-force attacks and is quantified:

$$|\mathcal{K}| = \prod_{i=1}^n |K(i)| = 2^{256}, \quad (23)$$

where $\mathcal{K}$ means the valid key set, n presents the key length and $n = 256$ in this work, $K(i)$ denotes the set of possible values for the i -th key component and $K(i) \in \{0, 1\}$. For a symmetric algorithm, the research [36] has shown that when the key space exceed 2^{100} , brute-force attacks become practically infeasible for adversaries. In this work, the key space is 2^{256} as shown Eq. (23), for exceeding this threshold, ensuring security.

2) Derivation Security Analysis

In this work, chained encryption at both block and channel levels is achieved through derived keys to realize a one-time-pad effect. Thus, the performance (randomness) of key derivation is critical to security. It is tested to the impact of different input changes on the derived key:

① *Different Images*. As shown in Fig. 9, even with identical $K_{1,1,1}$, the first derived key $K_{1,2,1}$ becomes entirely distinct across images. Similarly, channel-level keys $K_{1,1,2}$ and $K_{1,1,3}$ diverge completely. This confirms that the encryption process adapts substitution/permutation patterns uniquely to each image's content.

② *Varying Block Sizes*. Fig. 10 shows derived keys diverge immediately under different block sizes. This demonstrates that block size variations alone alter key derivation significantly, producing distinct encryption behaviors.

③ *Single-Pixel Modification*. one pixel in I_1 is modified, setting $B_{1,1,1}(1, 1) = 0$, to create I'_1 . Then, Encryption of both images yields entirely distinct derived keys as shown in Fig. 11. Notably, the first derived key $K_{1,2,1}$ differs completely despite minimal input changes.

④ *1-Bit Key Difference*. Encrypting an image with two keys differing by 1 bit $K'_{1,1,1} = K_{1,1,1} \oplus (\{0\}^{255} \cup \{1\})$ produces entirely distinct derived keys as shown in Fig. 12.

All experiments show derived keys diverge completely under parameter modifications, despite fixed $K_{1,1,1}$. This validates the cryptographic security in key derivation, achieving a one-time-pad effect without requiring fresh keys per image or block size.

3) Key Sensitivity Analysis

Key sensitivity measures how cryptographic outputs change in response to minor key modifications. A good encryption

algorithm must ensure that even a single-bit key alteration produces completely different encryption/decryption results. To evaluate this property, we encrypted image I_1 using key $K_{1,1,1}$ and then attempted decryption with: 1) the original key $K_{1,1,1}$; 2) a modified version $K'_{1,1,1}$ differing by just one bit $K'_{1,1,1} = K_{1,1,1} \oplus (\{0\}^{255} \cup \{1\})$.

As demonstrated in Fig. 13, decryption with $K'_{1,1,1}$ yields output visually indistinguishable from the encrypted image, where no meaningful information is recovered beyond what the ciphertext already contains. This demonstrates that key error can disrupt the decryption process. It should be noted that in this architecture, encrypting the same image with $K_{1,1,1}$ and $K'_{1,1,1}$ respectively produces completely different results, which are not presented.

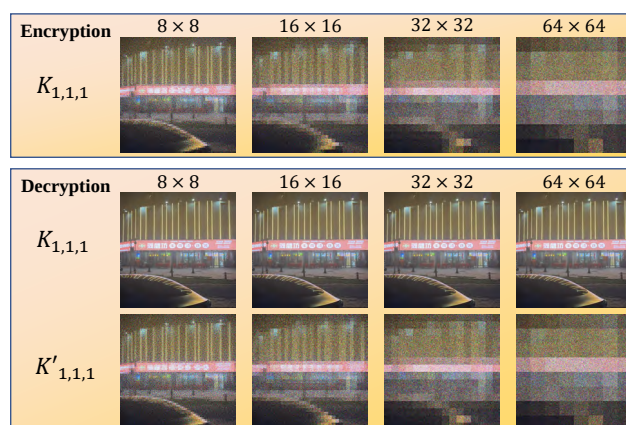


Fig. 13: An example of the key sensitivity.

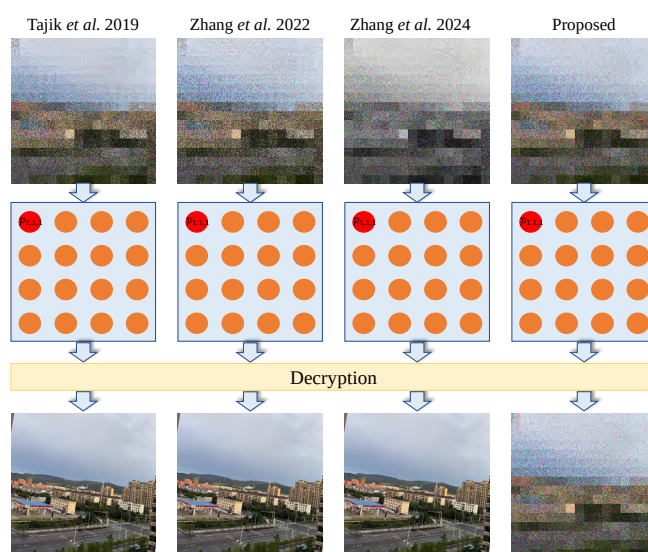


Fig. 14: Comparison examples with other representative TPE schemes (published by Tajik *et al.* 2019 [16], Zhang *et al.* 2022 [15], and Zhang *et al.* 2024 [37]) in sensitivity analysis ($p_{1,1,1} = 0$).

C. Comparisons with Previous Work

This part presents three comparative analyses with previous state-of-the-art work: sensitivity analysis, differential attack

analysis, and quantitative analysis. The first two experiments validate the security of the proposed architecture, while the quantitative analysis is used for evaluating result generalizability.

1) Sensitivity Analysis

Previous TPE schemes process pixels, blocks, and channels independently, creating encrypted images with excessive robustness. While this allows decrypted images to maintain visual fidelity despite significant modifications in encrypted images, such error tolerance constitutes a fundamental cryptographic weakness. This vulnerability enables practical perturbation attacks, particularly differential fault analysis (DFA) [38]. By intentionally modifying specific encrypted pixels/blocks, even minimally, an attacker could probe the encryption scheme and gradually deduce critical key information or recover partial plaintext data through analysis of the decrypted outputs. The proposed architecture overcomes this flaw, while maintaining visual effects, it ensures that manipulation in encrypted images can disrupt decryption. The following experiments verify that the proposed architecture exhibits sensitivity, while previous schemes do not.

Three representative previous TPE schemes [15], [16], [37] are selected: Tajik *et al.* 2019 [16] adopts the identical architecture as [32]; Zhang *et al.* 2022 [15] shares the same framework as other schemes based on data hiding; and Zhang *et al.* 2024 [37] aligns with the cross-plane encryption architecture. The image I_2 is selected and encrypted using each of these representative and the proposed schemes, as illustrated in Fig. 14. Subsequently, the first pixel value of the first channel $p_{1,1,1}$ in the encrypted image is modified, followed by decryption.

As shown in Fig. 14, the comparative experiments reveal critical security distinctions between previous schemes and the proposed architecture. When subjected to intentional ciphertext modification, existing TPE schemes successfully decrypt the tampered data, producing outputs visually identical to the original image. This error tolerance exposes a severe cryptographic weakness. Indeed, it enables potential attacks about DFA. In contrast, the proposed scheme exhibits strict sensitivity to ciphertext modifications. A slight modification causes complete decryption failure, rendering the output indistinguishable from the encrypted image, i.e., the decrypted result reveals no additional information beyond what is already contained in the encrypted image (zero information leakage).

Furthermore, to objectively access the sensitivity, it is employed to four standardized image metrics through comparative analysis with the original image I_2 : number of pixels change rate (NPCR, range: $[0,1]$), unified average changing intensity (UACI, range: $[0,1]$), peak signal-to-noise ratio (PSNR in dB, range: $[0, +\infty)$), and structural similarity index measure (SSIM, range: $[0,1]$). For NPCR, higher values demonstrate greater sensitivity to perturbations. For UACI, values closer to 0.33 indicate better sensitivity performance. For PSNR and SSIM, lower values indicate better cryptographic properties.

The comparative evaluation results in Table I reveals difference in sensitivity performance between existing TPE schemes and the proposed architecture. The three existing schemes demonstrate poor sensitivity to modifications, as evidenced by their lower NPCR and UACI values combined high PSNR

TABLE I: Objective indicator of encrypted image sensitivity analysis.

	Tajik <i>et al.</i> 2019 [16]	Zhang <i>et al.</i> 2022 [15]	Zhang <i>et al.</i> 2024 [37]	Proposed
NPCR	2.543×10^{-6}	0.046	1.271×10^{-6}	0.992
UACI	4.637×10^{-7}	1.799×10^{-4}	9.275×10^{-7}	0.198
PSNR	70.213	61.449	61.697	11.800
SSIM	0.999	0.999	0.999	0.077

TABLE II: Objective indicator of the differential attack.

	Tajik <i>et al.</i> 2019 [16]	Zhang <i>et al.</i> 2022 [15]	Zhang <i>et al.</i> 2024 [37]	Proposed
NPCR	0	2.543×10^{-6}	1.272×10^{-6}	0.975
UACI	0	7.879×10^{-7}	5.286×10^{-7}	0.208
PSNR	$+\infty$	64.511	66.581	10.633
SSIM	1	0.999	0.999	0.419

and SSIM scores. These metrics collectively indicate that they maintain nearly perfect visual fidelity in decrypted images even when the encrypted data has been altered. In addition, while Zhang *et al.* 2022 [15] shows slightly elevated NPCR/UACI values compared to Tajik *et al.* 2019 [16] and Zhang *et al.* 2024 [37], it stems from its inherent lossy compression nature rather than cryptographic sensitivity, as the distortion is irreversible and unrelated to security mechanisms.

In contrast, the proposed architecture exhibits fundamentally stronger security characteristics, with higher NPCR and UACI values, coupled with extremely low PSNR and SSIM scores. These results demonstrate that it achieves complete decryption failure when processing modified one pixel, fulfilling the critical requirement of cryptographic sensitivity. The experimental data demonstrate that the proposed architecture is a significant advancement over existing TPE schemes about sensitivity.

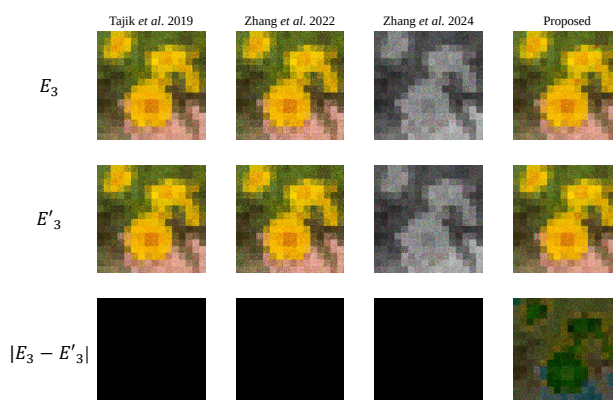


Fig. 15: Comparison examples with previous TPE schemes (published by Tajik *et al.* 2019 [16], Zhang *et al.* 2022 [15], and Zhang *et al.* 2024 [37]) in the differential attack.

2) Differential Attack Analysis

Differential cryptanalysis represents a fundamental method for assessing encryption security. It examines the propagation of differences by comparing encrypted images generated from

TABLE III: Average PSNR and SSIM values of the quantitative analysis.

Block Size		8	16	32	64
Tajik <i>et al.</i> 2019 [16]	PSNR	$+\infty$	$+\infty$	$+\infty$	$+\infty$
	SSIM	1	1	1	1
Zhang <i>et al.</i> 2022 [15]	PSNR	66.473	67.954	69.203	$+\infty$
	SSIM	0.999	0.999	0.999	1
Zhang <i>et al.</i> 2024 [37]	PSNR	19.627	19.887	20.124	20.501
	SSIM	0.431	0.522	0.633	0.725
Proposed	PSNR	$+\infty$	$+\infty$	$+\infty$	$+\infty$
	SSIM	1	1	1	1

nearly identical images, typically differing by just one pixel. A good encryption scheme should exhibit the avalanche effect, where minimal input variations produce completely dissimilar encrypted images.

The execution steps of this analysis are:

- 1) The original image I_3 , shown in Fig. 8, is selected, as illustrated in Fig. 15;
- 2) One modified version I'_3 is created by zeroing pixel $p_{1,1,1}$;
- 3) The corresponding encrypted images E_3 and E'_3 are generated through encryption;
- 4) The absolute differential image $|E_3 - E'_3|$ is computed.

This controlled experiment directly evaluates the resistance to differential attacks by quantifying how single-pixel perturbations propagate through the encryption process.

The experimental results shown in Fig. 15 reveal fundamental security differences. The previous schemes [15], [16], [37] produce identical encrypted images (evidenced by completely black differential images) when processing minimally different images, indicating failure to propagate input variations. On the other hand, the proposed architecture generates significantly distinct encrypted images, with the differential image demonstrating comprehensive pixel-level changes. This visual evidence confirms the proposed architecture superior ability to amplify minimal input differences through the encryption process.

Objective metric analysis in Table II reinforces these findings through four standardized metrics. While existing schemes show negligible differences, the proposed architecture alters nearly all pixel values (NPCR value > 0.97). The metrics thus demonstrate the proposed architecture successfully achieves cryptographic sensitivity.

3) Quantitative Analysis

While previous experiments focused on single-image case studies, for this experiment, we conduct large-scale quantitative analysis using the first 1000 images from the Helen dataset [39]. This systematic evaluation specifically assesses thumbnail preservation quality through two complementary metrics: PSNR and SSIM, which measure the visual quality between original and encrypted thumbnails. Higher values for both metrics indicate better preservation of the original thumbnail structural and visual characteristics in the encrypted

version, which is crucial for practical applications requiring visual browsing of encrypted images.

Table III presents the average PSNR and SSIM values across all test images. The results reveal that both Tajik *et al.* 2019 [16] and the proposed architecture achieve perfect thumbnail preservation. While Zhang *et al.* 2022 [15] demonstrates competitive performance, its approximate nature leads to minor but measurable deviations from exact thumbnail preservation. Additionally, as expected, Zhang *et al.* 2024 [37] yields the lowest values in both metrics, which can be visually confirmed, as shown in Fig. 15, since it converts color images to grayscale.

The experimental results conclusively demonstrate that the proposed architecture successfully maintains excellent thumbnail quality. This dual capability of visual preservation and security represents a significant advancement over previous work, and the block size dependency further provides system designers with a tunable parameter to optimize the balance between security and visual effects for specific use cases.

V. CONCLUSION

TPE is proposed to balance visual usability and privacy by retaining partial visual information, addressing the lack of visual accessibility in traditional encryption. However, existing TPE schemes perform fully independent encryption processes, where pixels, blocks, and channels remain mutually isolated, resulting in excessive robustness that compromises cryptographic security. In this work, we proposed a novel triple-chain TPE architecture that introduces cryptographic dependencies across pixel, block, and channel levels. By chaining encryption operations, where each output affects subsequent inputs, the architecture achieves unprecedented sensitivity: minimal changes in either original or encrypted images propagate globally, ensuring security while preserving visual usability. Finally, experimental results validate the proposed architecture’s sensitivity while demonstrating preserved visual quality and usability. In the future, researchers may explore theoretical security proofs and usability enhancements (beyond visual usability) to achieve a more trade-offed between usability and privacy, but this would likely require more advanced mathematical knowledge and profound cryptographic expertise.

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